

Maximum Index

Difficulty: Medium | Accuracy: 24.5% | Submissions: 381K+ | Points: 4

Given an array `arr[]` of positive integers, return the maximum difference between two indices `j` and `i` (i.e., $j - i$) such that $arr[i] \leq arr[j]$ and $i \leq j$.

Examples

Input: `arr[] = [1, 10]`

Output: 1

Explanation: $arr[0] \leq arr[1]$ so $(j-i)$ is $1-0 = 1$.

Input: `arr[] = [5, 4, 3]`

Output: 0

Explanation: There is no pair that satisfies the given condition.

Input: `arr[] = [34, 8, 10, 3, 2, 80, 30, 33, 1]`

Output: 6

Explanation: In the given array $arr[1] < arr[7]$ satisfying the required condition ($arr[i] \leq arr[j]$) thus giving the maximum difference of $j - i$ which is 6 (7-1).

Constraints

$1 \leq arr.size \leq 10^5$

$0 \leq arr[i] \leq 10^9$

Topic Tags

Arrays, Two-Pointer Algorithm

Company Tags

VMware, Amazon, Microsoft, MakeMyTrip, Google, Snapdeal